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Studierichting: Informatica
Oefeningenreeks 1C nr. 12.4

$z_1 = \sqrt{2} + 0i$

$z_2 = -1 + i$

Modulus:

$|z_1| = \sqrt{2}$

$|z_2| = \sqrt{2}$

Argumenten:

$\tan\theta_1 = 0$

$\theta_1 = \text{Bgtan}(0) = 0$

$\tan\theta_2 = -1$

$\theta_2 = \text{Bgtan}(-1) = -\frac{\pi}{4}$

$\theta_2 = -\frac{\pi}{4} + \pi = \frac{3\pi}{4}$

$\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2$

$\arg\left(\frac{z_1}{z_2}\right) = 0 - \frac{3\pi}{4} = -\frac{3\pi}{4}$

Goniometrische vorm:

$\frac{z_1}{z_2} = \cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)$

$\frac{z_1}{z_2} = \text{cis}\left(-\frac{3\pi}{4}\right)$

References